The Impossibility of Objective Uniqueness in Neoliberal Evaluation Models: A Mathematical Proof of Self-Destruction in the 1-Dimensional Projection of Non-Independent High-Dimensional Elements

Is what your conclusion?

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Abstract

This paper mathematically deconstructs the structural vulnerability of quantitative human and academic evaluation systems driven by contemporary neoliberalism and pervasive bureaucratization. We analyze this phenomenon through the multi-layered lenses of differential geometry, discrete mathematics, game theory, and infinite-dimensional linear algebra. We prove that any evaluation function projecting a high-dimensional vector of non-independent human values onto a 1-dimensional scalar inherently forfeits uniqueness and objectivity, thereby necessitating subjective intervention. Furthermore, we demonstrate that any deceptive model attempting to reject continuity or reduce infinite dimensions inevitably converges toward mathematical thermal death and self-destructive phase transitions—the ultimate demise of intellect.

1 Introduction: The Axiomatic Deception of Quantification

In modern society, particularly within higher education and research institutions, bureaucratic evaluation systems run rampant under the neoliberal governance paradigm. These systems attempt to uniformly manage the intellect and achievements of researchers via single numerical metrics (scalar values), such as the number of papers or citation counts. Although these systems claim to achieve "objective and efficient optimization," they fail to function as valid mathematical models. This paper provides a rigorous proof of the inescapable self-contradictions logically embedded within these quantitative evaluation models.

2 Geometric Proof of All-Dimensional Collapse via the Inner Product Model

Let the essence of human intellect and intrinsic value be represented as an n-dimensional vector $\boldsymbol{x} = (x_1, x_2, \ldots, x_n)$, where x_1 denotes the "number of papers" (visible output) and x_2 represents the "rigor and quality of research." Since any evaluation system that ranks and orders human beings on a 1-dimensional line mathematically defines a mapping, it fundamentally reduces to a local inner product (projection) with an evaluation criteria vector $\boldsymbol{w}$:

$$\text{Evaluation Score} = \boldsymbol{w} \cdot \boldsymbol{x} = |\boldsymbol{w}||\boldsymbol{x}| \cos\theta \tag{1}$$

Based on the geometric correlation (non-independence) between the constituent elements x_i and x_j, we prove that the evaluation model inherently collapses under all of the following three scenarios.

2.1 Pattern A: Anti-Interference (Negative Correlation / Trade-off: $\cos\theta < 0$)

The ultimate pursuit of "rigor (x_2)" logically and physically constrains the short-term "number of papers (x_1)." The moment the evaluator fixes the direction of $\boldsymbol{w}$ along the dimension of x_1, rational agents executing optimization behavior for mere survival will force x_2 to collapse to zero, thereby annihilating the qualitative essence of academia.

2.2 Pattern B: Interference (Positive Correlation / Dependence: $\cos\theta > 0$)

When elements are structurally interdependent, attempting to maximize a specific evaluation metric triggers systemic overfitting. This drives a hyper-inflation of hollow metrics, stripping the numerical values of any capacity to serve as reliable signals for measuring true worth.

2.3 Pattern C: Independence (Orthogonality: $\cos\theta = 0$)

When intrinsically unrelated elements (e.g., depth of erudition and political sycophancy) are orthogonal, the slightest tilt of the evaluation vector $\boldsymbol{w}$ toward the political dimension instantly renders the orthogonal dimension of erudition completely non-existent within the score. Consequently, invaluable intellect is reduced to zero and weaponized as a pretext for budget cuts.

3 Proofs of Complete Breakdown across Plurality, Discontinuity, and Infinite Dimensions

3.1 Manifold Theory and Rank Deficiency of the Jacobian Matrix

Let the space of human value be a non-linear, differentiable manifold M, and define the evaluation system as a differentiable mapping $f : M \to \mathbb{R}$. When elements are non-independent—implying that the space possesses curvature—the gradient vector $\nabla f(\boldsymbol{x})$ becomes a non-constant function dependent on the position $\boldsymbol{x}$.

To completely dismantle the evaluator's final defense that "fair quantification and comparison are possible within local, restricted domains," we define the following lemma:

Lemma 1 (Lemma of Local Non-Isometry)**.** *Let $\boldsymbol{x}_A \neq \boldsymbol{x}_B$ be two distinct points on the manifold M (representing individuals with different initial conditions or environments). When an identical effort/direction vector $\boldsymbol{d}$ is applied to both, the resulting increment in their evaluation scores (the primary vector evaluation variance) mathematically can never be equal:*

$$\nabla f(\boldsymbol{x}_A) \cdot \boldsymbol{d} \neq \nabla f(\boldsymbol{x}_B) \cdot \boldsymbol{d} \tag{2}$$

Thus, even local evaluation fairness is fundamentally and entirely lost in a non-independent high-dimensional space.

Furthermore, we determine the inevitable destruction of intrinsic value caused by rational agents attempting to maximize 1-dimensional numerical targets:

Lemma 2 (Lemma on Overfitting of Hacking Behavior)**.** *Let $t \to \infty$ be the number of optimization steps taken by evaluated agents attempting to maximize the scalar function $f(\boldsymbol{x})$ set by the evaluator. The value of the intrinsic dimension x_{true} (rigor, erudition), which stands in a non-independent trade-off relationship, deteriorates exponentially as the evaluation function converges to its boundary values:*

$$\lim_{t \to \infty} x_{true}(t) \exp(\lambda t) = 0 \quad (\lambda > 0) \tag{3}$$

In other words, a managed society that forces individuals to chase numbers mathematically guarantees a dystopia where the core of human value is forcibly compressed into its shallowest possible form.

3.2 NP-Hardness and Arrow's Impossibility Theorem in Discrete Spaces

If evaluators attempt to evade the continuous model by claiming that "the subjects of evaluation are discrete data, not a continuous manifold," the system transforms into a discrete optimization problem on a finite discrete graph

$G = (V, E)$. The optimization of non-independent discrete elements instantly gets trapped in the prison of **NP-hardness** (combinatorial explosion), requiring more than the lifespan of the universe to compute an optimal solution. Furthermore, according to **Arrow's Impossibility Theorem**, a function that consistently and without contradiction derives a 1-dimensional ranking for three or more alternatives based on multiple criteria is mathematically non-existent. The alleged "objectivity" of their evaluation is tissue-thin, reduced to nothing more than the violence of raw, dictatorial subjectivity.

3.3 The Convergence of Numerical Signals to White Noise in Imperfect Information Games

In a non-cooperative game between evaluators and evaluated agents, where the evaluator covertly alters parameters (the rules) and agents hack the scores, the unique Nash equilibrium achieved through infinite iterations dictates that the information content of the numerical metrics converges to absolute zero (**absolute white noise**). Relying on these scores for resource allocation automatically breeds an adverse selection loop, funnelling national wealth directly to the most proficient frauds while suffocating genuine intellect.

3.4 Loss of Uniqueness and Forced Subjective Intervention in Infinite-Dimensional Hilbert Spaces

Consider a bounded linear operator $T : \mathcal{H} \to \mathbb{R}^m$ mapping an infinite-dimensional Hilbert space $\mathcal{H}$ of human potential onto a finite-dimensional evaluation value $\mathbb{R}^m$. By the rank-nullity theorem, the null space ($\ker T$)—the dimensions completely erased and ignored by the evaluation—remains inherently infinite-dimensional. Moreover, the very definition of the inner product required to extract principal components from an infinite-dimensional space is 100% dependent on the evaluator's subjective weight function. The facade of "objective data" is entirely stripped away.

4 Conclusion: Condemning the Absence of Intrinsic Variables

Should the evaluators argue that "we evaluate fairly using a vast but finite set of multiple vectors," we must instantly retaliate with a single question:

"If so, why does your finite evaluation function fail to integrate the true essence of academia—the variable of rigorous content—and instead inflate only the shallow dimension of volume (the number of papers)?"

This structural distortion is irrefutable mathematical proof that their model does not function as a scientific tool, but operates purely as a puppet for political propaganda. We hereby completely reject the comprehensive violence of flattening high-dimensional intellect into 1 dimension.

5 Epilogue: A Warning from the Radical Academic Supremacists Who Love the Beauty of Mathematics and Detest Political Academism

We fully recognize that this stance may be deemed our own ego. We are well aware that historically, competition has driven academic progress. However, is competition truly an absolute necessity in our contemporary era? We intend to reflect deeply on this question, weighing it against the very arguments we have articulated here. Yet, let this never be forgotten: neoliberalism is nothing more than a subjective construct, and to force its compliance upon others is an insult to human dignity. To the politicians: We have cast our stone. Do not dare desecrate academia.